Strategies for Building an Operational Excellence Program

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INTRODUCTION

This study identifies best practices for building an OpEx program using insights from academic literature and a roundtable discussion with industry experts Operational Excellence (OpEx) enhances productivity, reduces costs, and improves efficiency. Continuous Improvement (CI) is central to OpEx, enabling organizations to refine operations and stay competitive.. AI-assisted analysis highlights key success factors and a structured framework for sustainable CI initiatives. The findings are published in ICAMS Technical Report 24-01.

BACKGROUND

- CI methodologies (Lean, Six Sigma, Toyota Production System) focus on waste reduction, process standardization, and efficiency.
- OpEx goes beyond tools, requiring leadership commitment, employee engagement, and strategic alignment.
- Organizations must integrate these principles into intentional programs for long-term sustainability and competitive advantage.

There is a need to synthesize academic research on CI/OpEx into information easily accessible to industry.

METHODOLOGY

- **Literature Review**: Identified key success factors, challenges, and frameworks for CI programs.
- **Roundtable Discussion**: Gathered insights from industry experts on OpEx implementation.
- **AI-Assisted Analysis**: Extracted themes, success factors, and barriers from roundtable transcripts.
- **Framework Development**: Synthesized findings into a structured, five-phase OpEx model.

Key Characteristics of Successful Continuous Improvement Programs with Success Factors

Leadership	❖Respect for Individuals ❖Active Involvement ❖Clear Objectives
Employee Involvement	❖Individual Ownership ❖Accountability ❖Commitment
Performance Assessment	❖Use of Analytical Tools ❖Determination of Maturity Levels ❖Benchmarking
Organizational Communication/Structure	❖Alignment ❖Cross-Functional Collaboration ❖Transparency
Proper Training/Learning	❖Standardized Training ❖Continuous Assessment of Training
Standardization	❖Clarity ❖Standardization Involves Continuous Improvement ❖Consistency ❖Single Approach/Terminology
Integration of Technology	❖Value Addition ❖Readiness ❖Digitalization
Focus on Customer Satisfaction	❖Performance Measures Based on Strategic Objectives ❖Understanding Customer Expectations ❖Cost Effective Services
Longevity and Sustainment	❖Adaptability ❖Consideration of Social and Economic Environments ❖Promotion of Employee Interests

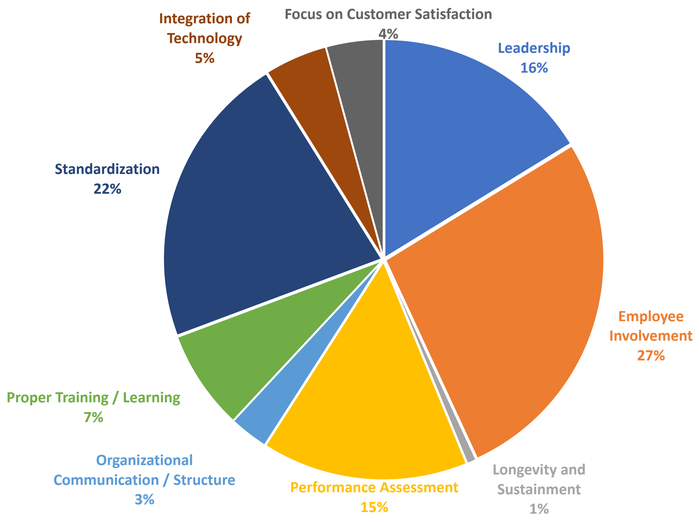
RESULTS

Key Elements of a C.I. program referenced on articles (Lit. Review)

Key Element	Literature Reference	%
Leadership	10	21%
Employee Involvement	5	11%
Performance Assessment	4	9%
Organizational Communication / Structure	5	11%
Proper Training / Learning	6	13%
Standardization	6	13%
Integration of Technology	4	9%
Focus on Customer Satisfaction	5	11%
Longevity and Sustainment	2	4%

"Leadership should replace supervision. Leadership is to help people to do a better job."

Figure 1: Consolidated Key Elements from Roundtable References



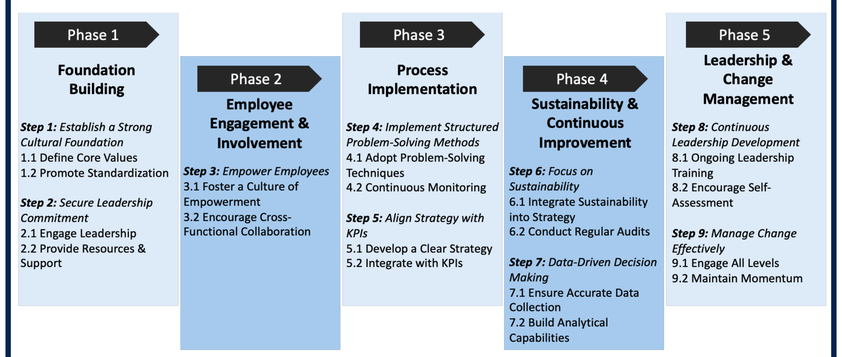
From Figure 1, we can observe that **employes involvement**, **standardization** and **leadership** consistently appears as a critical factor, underscoring its importance in achieving operational excellence.

"The communication between leadership and the floor has to be tight because if you're not tight, nothing's going to work."

RESULTS (cont.)



Experts from the automotive and manufacturing industries discussing best practices in Continuous Improvement and Operational Excellence at Auburn University (April 24, 2024)



This flowchart illustrates a step-by-step framework for implementing and sustaining a OpEx program, encapsulating all critical phases and activities.

FUTURE WORK

- Investigate existing **Systematic Literature Reviews (SLRs)** for similar themes.
- Conduct a deeper examination of **additional literature and case studies** on continuous improvement and operational excellence.
- Validate and refine the **current framework** through expanded research.
- Develop a **journal publication** to consolidate findings and provide practical guidance for organizations enhancing their OpEx initiatives.

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